Plane Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Triangle angle sum and exterior angles**

A triangle has angles measuring x, $2x$, and $3x$. Find each angle.

$x + 2x + 3x = 180$, so $6x = 180$, giving $x = 30$. Angles: 30°, 60°, 90°.

2 **[Foundational]** • **Target: Area and perimeter of polygons**

Find the area of a triangle with base 14 and height 9.

Area $= \frac{1}{2}(14)(9) = 63$.

3 **[Foundational]** • **Target: Area and perimeter of polygons**

Find the area of a parallelogram with base 12 and height 5.

Area $=$ base $\times$ height $= 12 \times 5 = 60$.

4 **[Foundational]** • **Target: Area and perimeter of polygons**

Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.

$A = \frac{3\sqrt{3}}{2}(6)^2 = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3}$.

5 **[Foundational]** • **Target: The Pythagorean theorem and special right triangles**

A right triangle has legs 9 and 12. Find the hypotenuse.

$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

6 **[Foundational]** • **Target: The Pythagorean theorem and special right triangles**

A 45-45-90 triangle has a leg of length 8. Find the length of the hypotenuse.

In a 45-45-90 triangle, the hypotenuse equals leg $\times \sqrt{2}$: $8\sqrt{2}$.

7 **[Foundational]** • **Target: The Pythagorean theorem and special right triangles**

A 30-60-90 triangle has a hypotenuse of 20. Find the lengths of both legs.

In a 30-60-90 triangle, the shorter leg is $\frac{\text{hypotenuse}}{2}$ and the longer leg is $\frac{\text{hypotenuse}}{2} \times \sqrt{3}$: shorter leg $= 10$, longer leg $= 10\sqrt{3}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Circles: area, circumference, arcs, and sectors

Find the area and circumference of a circle with diameter 14, in terms of π.

Radius $= 7$. Area $= \pi r^2 = 49\pi$. Circumference $= \pi d = 14\pi$.

9 [Intermediate] • Target: Circles: area, circumference, arcs, and sectors

Find the arc length of a 120° sector in a circle with radius 9, in terms of π.

Arc length $= \frac{120}{360} \times 2\pi(9) = \frac{1}{3} \times 18\pi = 6\pi$.

10 [Intermediate] • Target: Volume and surface area of 3D solids

Find the volume of a rectangular prism with dimensions 5, 8, and 10.

Volume $= 5 \times 8 \times 10 = 400$.

11 [Intermediate] • Target: Volume and surface area of 3D solids

Find the volume of a cone with radius 6 and height 10, in terms of π.

Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$.

12 [Intermediate] • Target: Trapezoids and composite polygon areas

Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

Area $= \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66$.

13 [Intermediate] • Target: Trapezoids and composite polygon areas

An L-shaped figure is formed by a 10-by-8 rectangle with a 4-by-3 rectangle removed from one corner. Find the remaining area.

Large rectangle area: $10 \times 8 = 80$. Removed rectangle area: $4 \times 3 = 12$. Remaining area: $80 - 12 = 68$.

14 [Intermediate] • Target: Chords, tangents, and inscribed angles in circles

An inscribed angle subtends an arc of 100°. Find the measure of the inscribed angle.

An inscribed angle is always half the arc it subtends: $\frac{100^\circ}{2} = 50^\circ$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: A multi-part composite geometry problem • Cross-Lesson: 6.01 Trigonometry

This question has two parts. A semicircular window has a diameter of 12 feet and sits atop a rectangular window of width 12 feet and height 5 feet. (a) Find the area of the semicircular portion, in terms of π. (b) Find the total area of the combined window shape.

(a) Radius $= 6$. Semicircle area $= \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(36) = 18\pi$. (b) Rectangle area: $12 \times 5 = 60$. Total area: $60 + 18\pi$ square feet.

16 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.

$A = \frac{3\sqrt{3}}{2}(6)^2 = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3}$.

17 [Challenging] • Target: Angle relationships • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Two angles are supplementary, and one is 3 times the other. Find both angles.

Let the angles be x and $3x$: $x + 3x = 180$, so $4x = 180$, giving $x = 45$. The angles are 45° and 135°.

18 [Challenging] • Target: Angle relationships • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Two parallel lines are cut by a transversal, creating an angle of 70°. Find the measure of its alternate interior angle and its corresponding angle.

Alternate interior angles are equal: 70°. Corresponding angles are also equal: 70°.

19 [Challenging] • Target: Triangle angle sum and exterior angles • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A triangle's exterior angle at one vertex measures 115°. If one of the remote interior angles is 50°, find the other remote interior angle.

By the exterior angle theorem, the exterior angle equals the sum of the two remote interior angles: $115 = 50 + x$, so $x = 65^\circ$.

20 [Challenging] • Target: Similar and congruent figures • Cross-Lesson: 3.01 Functions & Graphs

Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.

Areas of similar figures scale by the SQUARE of the linear scale factor: $\left(\frac{5}{3}\right)^2 = \frac{25}{9}$. Larger triangle's area: $18 \times \frac{25}{9} = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Angle relationships

Two angles are complementary, and they differ by 52°. Find the larger angle.

a. 19°

b. 66°

c. 76°

d. 71° ✓

Let the angles be x and $x + 52$: $x + (x + 52) = 90$, so $2x = 38$, giving $x = 19$. The larger angle is $19 + 52 = 71^\circ$.

22 [Foundational] • Target: Triangle angle sum and exterior angles

Two angles are complementary, and they differ by 8°. Find the larger angle.

a. 49° ✓

b. 44°

c. 41°

d. 54°

Let the angles be x and $x + 8$: $x + (x + 8) = 90$, so $2x = 82$, giving $x = 41$. The larger angle is $41 + 8 = 49^\circ$.

23 [Foundational] • Target: Area and perimeter of polygons

Two angles are complementary, and they differ by 48°. Find the larger angle.

a. 74°

b. 64°

c. 21°

d. 69° ✓

Let the angles be x and $x + 48$: $x + (x + 48) = 90$, so $2x = 42$, giving $x = 21$. The larger angle is $21 + 48 = 69^\circ$.

24 [Foundational] • Target: The Pythagorean theorem and special right triangles

Two angles are supplementary, and they differ by 46°. Find the larger angle.

a. 67°

b. 108°

c. 113° ✓

d. 118°

Let the angles be x and $x + 46$: $x + (x + 46) = 180$, so $2x = 134$, giving $x = 67$. The larger angle is $67 + 46 = 113^\circ$.

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25 [Foundational] • Target: Circles: area, circumference, arcs, and sectors

Two angles are supplementary, and they differ by 30°. Find the larger angle.

a. 105° ✓

b. 100°

c. 75°

d. 110°

Let the angles be x and $x + 30$: $x + (x + 30) = 180$, so $2x = 150$, giving $x = 75$. The larger angle is $75 + 30 = 105^\circ$.

26 [Foundational] • Target: Similar and congruent figures

Two angles are complementary, and they differ by 22°. Find the larger angle.

a. 34°

b. 61°

c. 51°

d. 56° ✓

Let the angles be x and $x + 22$: $x + (x + 22) = 90$, so $2x = 68$, giving $x = 34$. The larger angle is $34 + 22 = 56^\circ$.

27 [Foundational] • Target: Volume and surface area of 3D solids

Two angles are complementary, and they differ by 76°. Find the larger angle.

a. 88°

b. 7°

c. 78°

d. 83° ✓

Let the angles be x and $x + 76$: $x + (x + 76) = 90$, so $2x = 14$, giving $x = 7$. The larger angle is $7 + 76 = 83^\circ$.

28 [Foundational] • Target: Multi-step geometry problems

Two angles are complementary, and they differ by 24°. Find the larger angle.

a. 57° ✓

b. 52°

c. 33°

d. 62°

Let the angles be x and $x + 24$: $x + (x + 24) = 90$, so $2x = 66$, giving $x = 33$. The larger angle is $33 + 24 = 57^\circ$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Trapezoids and composite polygon areas**

A right triangle has legs 16 and 30. Find the length of the hypotenuse.

a. 35

b. 34 ✓

c. 33

d. 46

$c = \sqrt{16^2 + 30^2} = \sqrt{256 + 900} = \sqrt{1156} = 34.$

30 **[Intermediate]** • **Target: Chords, tangents, and inscribed angles in circles**

A right triangle has legs 6 and 8. Find the length of the hypotenuse.

a. 9

b. 11

c. 14

d. 10 ✓

$c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$

31 **[Intermediate]** • **Target: Congruence and triangle proofs**

A right triangle has legs 9 and 40. Find the length of the hypotenuse.

a. 41 ✓

b. 42

c. 40

d. 49

$c = \sqrt{9^2 + 40^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41.$

32 **[Intermediate]** • **Target: A multi-part composite geometry problem**

A right triangle has legs 14 and 48. Find the length of the hypotenuse.

a. 50 ✓

b. 62

c. 49

d. 51

$c = \sqrt{14^2 + 48^2} = \sqrt{196 + 2304} = \sqrt{2500} = 50.$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A right triangle has legs 10 and 24. Find the length of the hypotenuse.

a. 27

b. 26 ✓

c. 25

d. 34

$c = \sqrt{10^2 + 24^2} = \sqrt{100 + 576} = \sqrt{676} = 26.$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has legs 18 and 24. Find the length of the hypotenuse.

a. 42

b. 31

c. 30 ✓

d. 29

$c = \sqrt{18^2 + 24^2} = \sqrt{324 + 576} = \sqrt{900} = 30.$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A right triangle has legs 20 and 48. Find the length of the hypotenuse.

a. 52 ✓

b. 68

c. 53

d. 51

$c = \sqrt{20^2 + 48^2} = \sqrt{400 + 2304} = \sqrt{2704} = 52.$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has radius 11. Find the arc length of a 270° sector, in terms of π.

a. $\frac{33}{2}\pi$ ✓

b. $\frac{35}{2}\pi$

c. $\frac{31}{2}\pi$

d. 33π

Arc length $= \frac{\theta}{360^\circ} \times 2\pi r = \frac{270}{360} \times 2\pi(11) = \frac{33}{2}\pi.$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A circle has radius 2. Find the sector area of a 180° sector, in terms of π.

a. 2π ✓

b. 3π

c. 4π

d. 1π

Sector area $= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{180}{360} \times \pi(4) = 2\pi.$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A circle has radius 12. Find the arc length of a 30° sector, in terms of π.

a. 1π

b. 3π

c. 2π ✓

d. 4π

Arc length $= \frac{\theta}{360^\circ} \times 2\pi r = \frac{30}{360} \times 2\pi(12) = 2\pi$.

39 [Challenging] • Target: Angle relationships • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the volume of a cylinder with radius 5 and height 14, in terms of π.

a. 350π ✓

b. 360π

c. 140π

d. 340π

Volume $= \pi r^2 h = \pi(25)(14) = 350\pi$.

40 [Challenging] • Target: Triangle angle sum and exterior angles • Cross-Lesson: 3.01 Functions & Graphs

Find the volume of a cone with radius 5 and height 10, in terms of π.

a. $\frac{244}{3}\pi$

b. $\frac{250}{3}\pi$ ✓

c. $\frac{256}{3}\pi$

d. 250π

Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(25)(10) = \frac{250}{3}\pi$.

41 [Challenging] • Target: Area and perimeter of polygons • Cross-Lesson: 4.01 Coordinate Geometry

Two similar triangles have a scale factor of $\frac{2}{5}$. If the smaller triangle has area 16, find the area of the larger triangle.

a. 105

b. 40

c. 95

d. 100 ✓

Areas of similar figures scale by the square of the linear scale factor: $\left(\frac{5}{2}\right)^2 = \frac{25}{4}$. Larger triangle's area: $16 \times 25/4 = 100$.

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42 [Challenging] • Target: The Pythagorean theorem and special right triangles • Cross-Lesson: 6.01 Trigonometry

Two similar triangles have a scale factor of $\frac{1}{2}$. If the smaller triangle has area 1, find the area of the larger triangle.

a. 1

b. 9

c. 2

d. 4 ✓

Areas of similar figures scale by the square of the linear scale factor: $\left(\frac{2}{1}\right)^2 = \frac{4}{1}$. Larger triangle's area: $1 \times 4/1 = 4$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Two similar triangles have a scale factor of $\frac{2}{5}$. If the smaller triangle has area 16, find the area of the larger triangle.

a. 105

b. 40

c. 95

d. 100 ✓

Areas of similar figures scale by the square of the linear scale factor: $\left(\frac{5}{2}\right)^2 = \frac{25}{4}$. Larger triangle's area: $16 \times 25/4 = 100$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Two parallel lines are intersected by a transversal. If one interior angle measures $(3x + 15)^\circ$ and its consecutive interior angle measures $(2x + 45)^\circ$, what is the value of x?

a. 24 ✓

b. 30

c. 36

d. 40

Consecutive interior angles are supplementary:
$(3x + 15) + (2x + 45) = 180 \Rightarrow 5x + 60 = 180 \Rightarrow 5x = 120 \Rightarrow x = 24$.

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

In a 30°-60°-90° right triangle, the length of the hypotenuse is $16\sqrt{3}$. What is the length of the side opposite the 60° angle?

a. 12

b. 24 ✓

c. $8\sqrt{3}$

d. 16

Side opposite 30° is half the hypotenuse: $\frac{16\sqrt{3}}{2} = 8\sqrt{3}$. Side opposite 60° is $(8\sqrt{3})\sqrt{3} = 8 \times 3 = 24$.

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